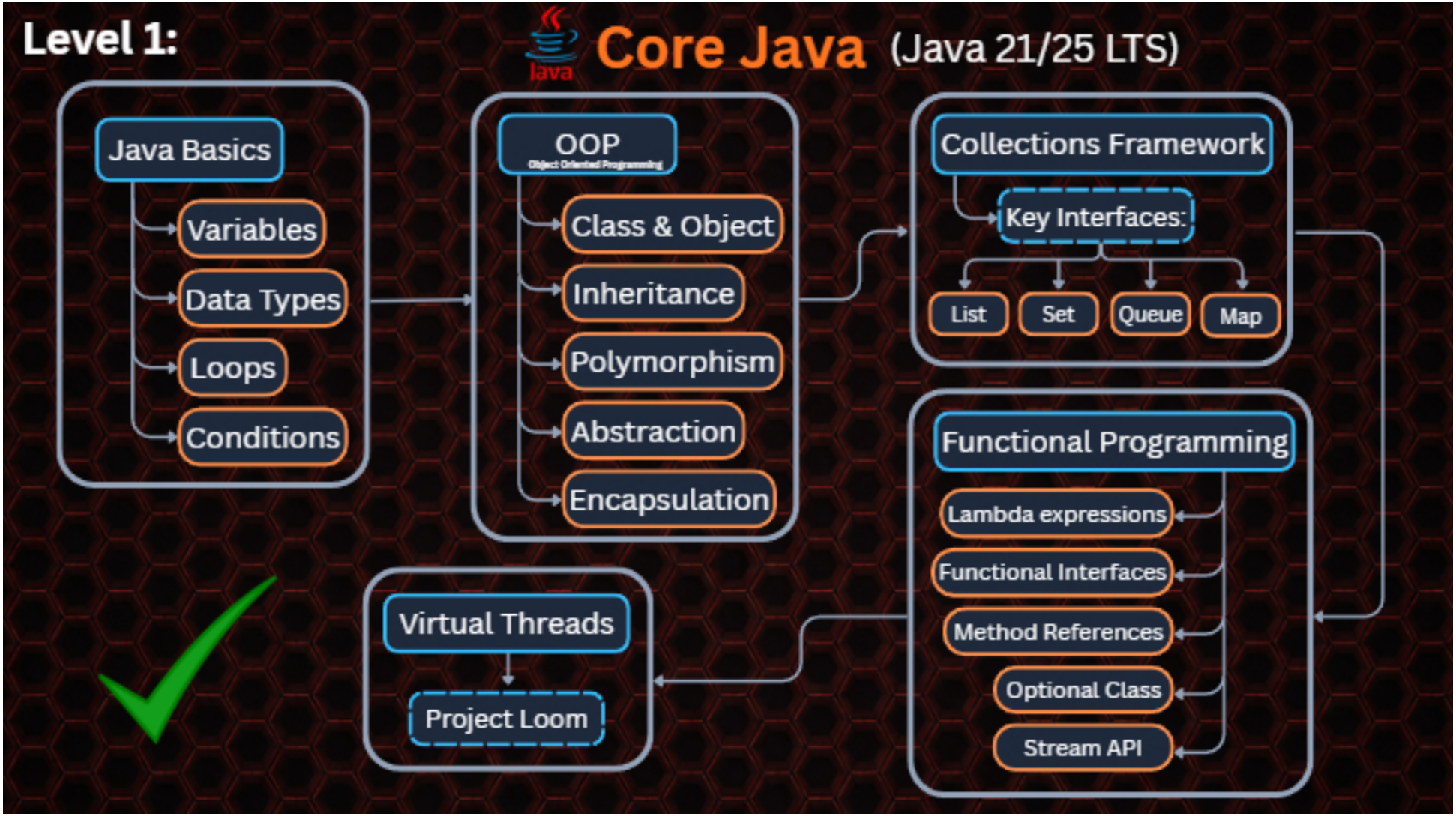

Level 1:
Core Java (Java 21/25 LTS)
Java Basics
Variables
Data Types
Loops
Conditions
OOP
Object Oriented Programming
Class & Object
Inheritance
Polymorphism
Abstraction
Encapsulation
Collections Framework
Key Interfaces:
List
Set
Queue
Map
Functional Programming
Lambda expressions
Functional Interfaces
Method References
Optional Class
Stream API
Virtual Threads
Project Loom

Level 2: 🛠️ Developer Essentials

Build Tools

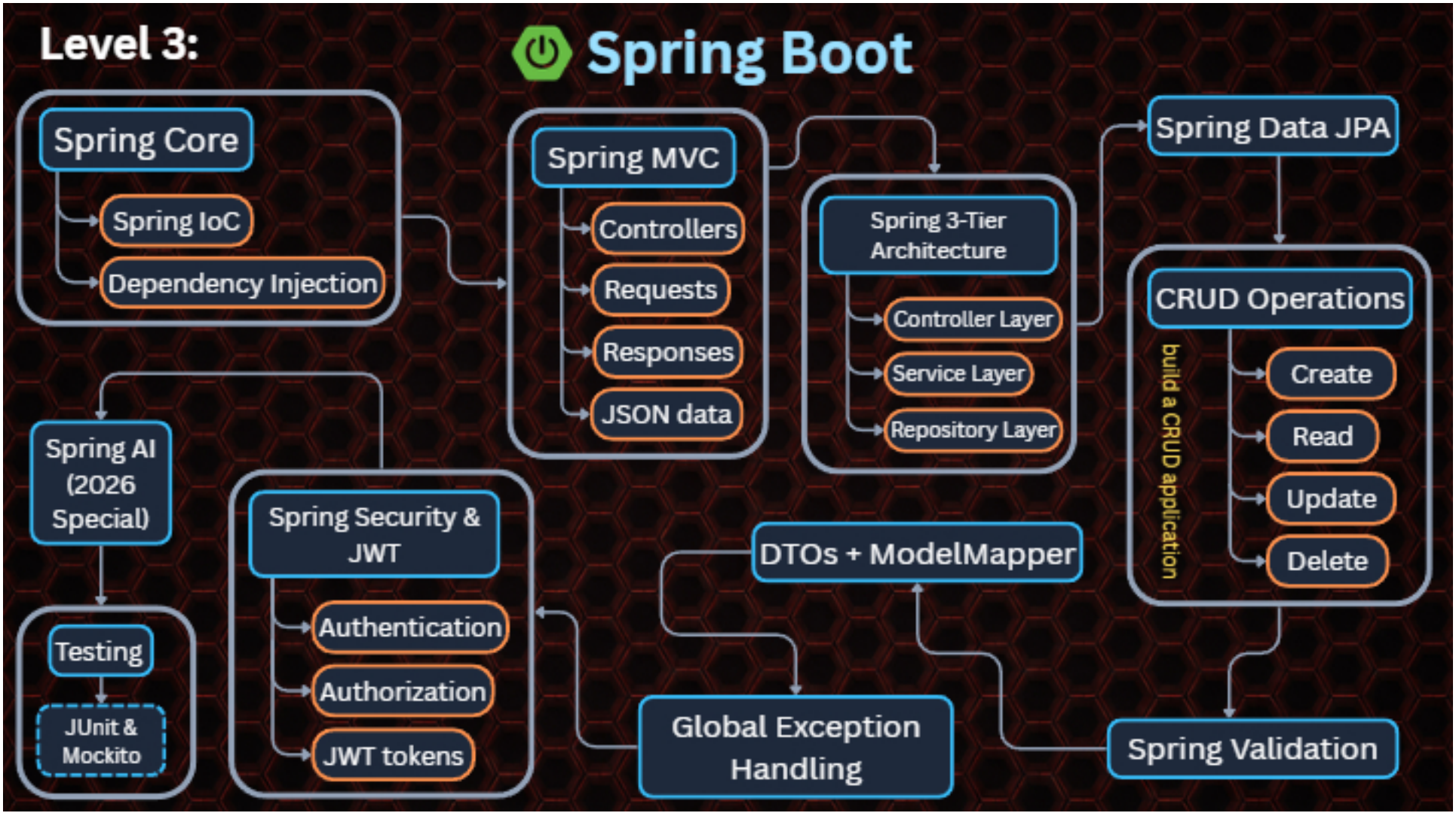
- Maven
- Gradle

- ✔ Download and Manage dependencies
- ✔ Manage versions
- ✔ Build your project

Git & GitHub

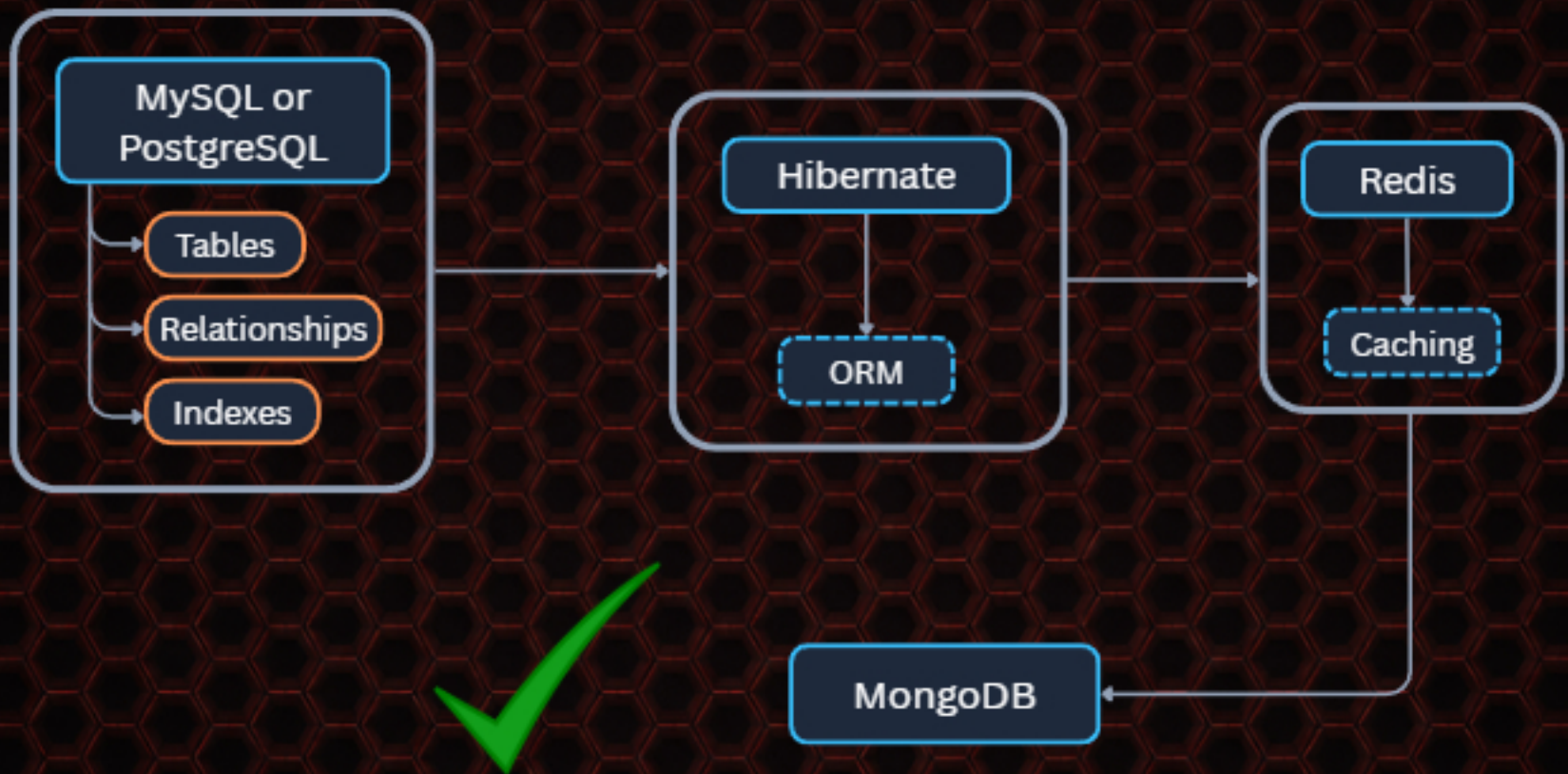
- commit
- push
- pull
- branching

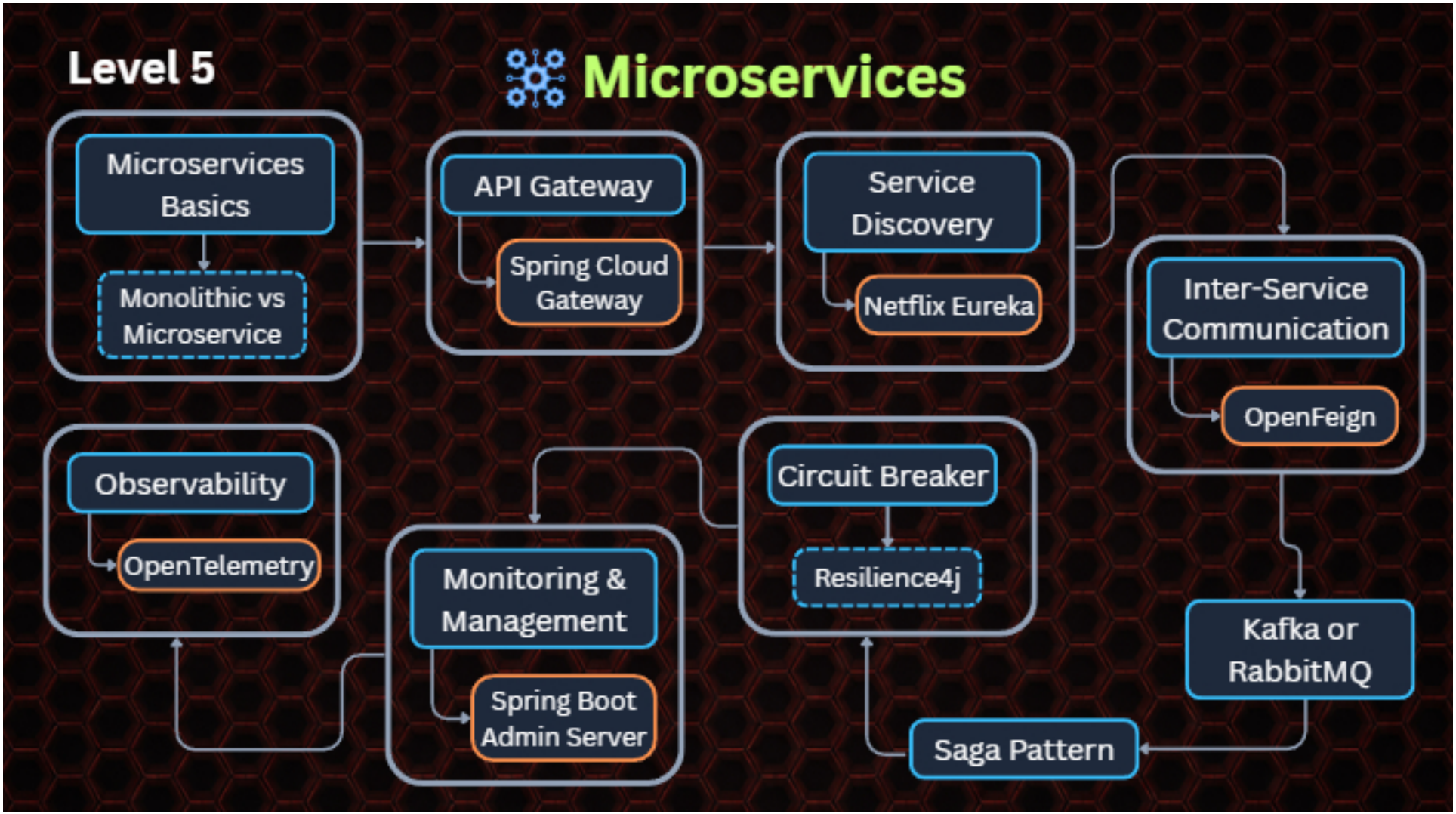
- ✔ Track changes
- ✔ Collaborate with teams
- ✔ Recover from mistakes


Level 3:
Spring Boot
Spring Core
Spring IoC
Dependency Injection
Spring MVC
Controllers
Requests
Responses
JSON data
Spring 3-Tier Architecture
Controller Layer
Service Layer
Repository Layer
Spring Data JPA
CRUD Operations
build a CRUD application
Create
Read
Update
Delete
Spring AI (2026 Special)
Testing
JUnit & Mockito
Spring Security & JWT
Authentication
Authorization
JWT tokens
DTOs + ModelMapper
Global Exception Handling
Spring Validation

Level 4

Database & Performance




Level 5
Microservices
Microservices Basics
Monolithic vs Microservice
API Gateway
Spring Cloud Gateway
Service Discovery
Netflix Eureka
Inter-Service Communication
OpenFeign
Observability
OpenTelemetry
Circuit Breaker
Resilience4j
Monitoring & Management
Spring Boot Admin Server
Kafka or RabbitMQ
Saga Pattern

Level 6

Devops & Cloud